

# Optimization Techniques

## Lab n°6 - Quadratic Programming

Consider a *Quadratic Programming* under *equality* constraints, i.e. an optimization problem of the form

$$\begin{aligned} &\text{minimize} && f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{x}^T \mathbf{c} \\ &\text{subject to} && A\mathbf{x} = \mathbf{b}, \end{aligned}$$

where  $Q$  is a symmetric matrix of dimension  $n$  and  $A$  is a (full-rank) matrix of dimension  $m \times n$ , with  $m < n$ .

To solve such a problem, one applies Newton-Raphson method to the Lagrangian

$$\mathcal{L}(\mathbf{x}, \lambda) = \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{x}^T \mathbf{c} + \lambda(A\mathbf{x} + \mathbf{b}).$$

As  $\mathcal{L}$  is a quadratic function, NR algorithm converges in *one step*, so that one can remove the while loop in the following pseudocode.

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**Algorithm 1** Newton-Raphson Method for Equality-Constrained QP

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**Input:** A positive definite matrix  $Q \in \mathbb{R}^{n \times n}$ , a cost vector  $\mathbf{c} \in \mathbb{R}^n$ , a constraint matrix  $A \in \mathbb{R}^{m \times n}$ , a right-hand side vector  $\mathbf{b} \in \mathbb{R}^m$ , an initial point  $\mathbf{x}_0$ .

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1:  $\mathbf{x} \leftarrow \mathbf{x}_0$ 
2: if  $\text{rank}(A) < m$  then                                 $\triangleright$  Check linear independence of constraints
3:   return "Linearly dependent constraints!"
4: break
5: end if
6:  $opt \leftarrow \text{False}$ 
7:  $k \leftarrow 0$ 
8:  $max\_iter \leftarrow 20$ 
9: while not  $opt$  and  $k < max\_iter$  do
    $H_{\mathcal{L}} \leftarrow \begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix}$                                  $\triangleright$  Hessian of the Lagrangian
    $\mathbf{g} \leftarrow \begin{bmatrix} Q\mathbf{x} + \mathbf{c} \\ A\mathbf{x} - \mathbf{b} \end{bmatrix}$                                  $\triangleright$  Gradient of the Lagrangian
    $H_{\mathcal{L}} \begin{bmatrix} \mathbf{p} \\ \lambda \end{bmatrix} = -\mathbf{g}$                                  $\triangleright$  Solve the linear system for step  $\mathbf{p}$  and multipliers  $\lambda$ 
10:  if  $\|\mathbf{p}\| \leq 10^{-7}$  then                                 $\triangleright$  Tolerance check for convergence
11:     $cch \leftarrow Q\mathbf{x} + \mathbf{c} + A^T \lambda$                                  $\triangleright$  Lagrangian stationarity check
12:    Print "Found minimum!"
13:     $opt \leftarrow \text{True}$ 
14:  else
15:     $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{p}$                                  $\triangleright$  Newton-Raphson update
16:  end if
17:   $k \leftarrow k + 1$ 
18: end while
19: return  $\mathbf{x}, \lambda$ 
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**Remark 1** The true gradient of the Lagrangian should be

$$\nabla \mathcal{L}(\mathbf{x}, \lambda) = \begin{pmatrix} Q\mathbf{x} + \mathbf{c} + \lambda A \\ A\mathbf{x} - \mathbf{b} \end{pmatrix}.$$

However, thanks to the linearity with respect to  $\lambda$ , one can compute directly the value of the multipliers and not its increment (as done, for instance, for  $\mathbf{p}$ , which is the increment of  $\mathbf{x}$ ).

**Exercise 1** Program Algorithm 1 and apply it to the following problem:

$$Q = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & -0.5 & 0 & 0 \\ 0 & -0.5 & 2 & 2 & 0.2 \\ 0 & 0 & 2 & 3 & 0 \\ 0 & 0 & 0.2 & 0 & 0.5 \end{pmatrix} \quad \mathbf{c} = (2, 1, 0, 0, -1)$$

$$A = \begin{pmatrix} 0 & 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix}.$$

Choose any starting point.

Verify that Lagrange's Theorem is satisfied at the endpoint.

Expected result:  $\mathbf{x}^* = (2.2487, -0.4974, -0.7534, 0.2540, 3.7513)$ ,  $\lambda^* = (-0.7250, -1.8857, 0.0200)$

**Exercise 2** Use the precedent program as a building block to implement SQP for the minimization of a function with equality constraints, in the nonlinear setting (see pseudocode in Algorithm 2).

Apply it to the following problem

$$\begin{aligned} \text{minimize} \quad & f(\mathbf{x}) = \frac{3}{2}x_2 - 2x_1^2 - x_1^3 + x_1^4 \\ \text{subject to} \quad & x_1^2 + x_2^2 = 3, \end{aligned}$$

starting from the point  $\mathbf{x}_0 = (1, 5)$ .

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#### Algorithm 2 Local Sequential Quadratic Programming (LSQP)

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**Input:** a cost  $f$ , constraints  $h$ , a starting point  $\mathbf{x}_{\text{start}}$ , a tolerance  $\epsilon$ .

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1: stop_crit  $\leftarrow 1$ 
2:  $k \leftarrow 0$ 
3:  $\mathbf{x}_0 \leftarrow \mathbf{x}_{\text{start}}$ 
4:  $\lambda_0 \leftarrow 0$  ▷ Initialize the Multipliers at 0
5: while stop_crit  $> \epsilon$  do
6:   Define Lagrangian:  $\mathcal{L}(\mathbf{x}) \leftarrow f(\mathbf{x}) + \lambda_0 \cdot h_{cs}(\mathbf{x})$ 
7:   Calculate objective gradient:  $\mathbf{g} \leftarrow \nabla f(\mathbf{x}_0)$ 
8:   Calculate Lagrangian Hessian:  $Q_m \leftarrow \nabla^2 \mathcal{L}(\mathbf{x}_0)$ 
9:   Calculate constraint Jacobian:  $A_m \leftarrow \nabla h_{cs}(\mathbf{x}_0)^T$ 
10:  Solve QP subproblem with Algorithm 1
    (with  $Q = Q_m$ ,  $A = A_m$  and  $\mathbf{g}$ ,  $h$  instead of
     $Q\mathbf{x} + \mathbf{c}$ ,  $A\mathbf{x} - \mathbf{b}$ , respectively)
    obtaining  $\mathbf{x}_{\text{new}}, \lambda_{\text{new}}$ 
11:  stop_crit  $\leftarrow \|\mathbf{x}_0 - \mathbf{x}_{\text{new}}\|$ 
12:   $\mathbf{x}_0 \leftarrow \mathbf{x}_{\text{new}}$ 
13:   $\lambda_0 \leftarrow \lambda_{\text{new}}$ 
14: end while
15: return  $\mathbf{x}_0$ 

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Expected result:  $\mathbf{x}^* = (1.2801, -1.1668)$ .

Consider now a *Quadratic Programming* under *inequality* constraints, i.e. an optimization problem of the form

$$\begin{aligned} &\text{minimize} && f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{x}^T \mathbf{c} \\ &\text{subject to} && A \mathbf{x} \leq \mathbf{b}, \end{aligned}$$

where  $Q$  is a symmetric matrix of dimension  $n$  and  $A$  is a (full-rank) matrix of dimension  $m \times n$ , with  $m < n$ .

Algorithm 3 details the pseudocode of the *Active Set Method*.

**Exercise 3** Program Algorithm 1 and apply it to the following problem:

$$\begin{aligned} Q &= \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} & \mathbf{c} &= (4, -5) \\ A &= \begin{pmatrix} -1 & 0 \\ 1 & 0 \\ 1 & -1 \\ 0 & 1 \end{pmatrix} & B &= \begin{pmatrix} 0 \\ 1 \\ 0 \\ 3 \end{pmatrix}. \end{aligned}$$

Choose the starting point  $\mathbf{x}_0 = (1, 1)$ .

Expected result:  $\mathbf{x}^* = (0, 2.5)$ .

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**Algorithm 3** Active Set Method for strictly convex Quadratic Programming

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**Input:** a positive definite cost matrix  $Q \in \mathbb{R}^{n \times n}$ , a cost vector  $c \in \mathbb{R}^n$ , a constraint matrix  $A \in \mathbb{R}^{m \times n}$ , a bounds vector  $b \in \mathbb{R}^m$ , an initial feasible point  $x_0$ , and an initial working set  $\mathcal{W}_0$ .

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1:  $x \leftarrow x_0$ 
2:  $\mathcal{W} \leftarrow \mathcal{W}_0$ 
3:  $\text{OptFound} \leftarrow \text{False}$ 
4: while not  $\text{OptFound}$  do
5:    $mac \leftarrow |\mathcal{W}|$  ▷ Number of active constraints
6:    $A_{\mathcal{W}} \leftarrow A(\mathcal{W}, :)$  ▷ Extract rows of  $A$  corresponding to active set
7:   Solve the KKT system for the search direction  $p$  and multipliers  $\lambda$ :
     
$$\begin{bmatrix} Q & A_{\mathcal{W}}^T \\ A_{\mathcal{W}} & 0 \end{bmatrix} \begin{bmatrix} p \\ \lambda \end{bmatrix} = \begin{bmatrix} -(Qx + c) \\ 0 \end{bmatrix}$$

8:   if  $\|p\| \leq \epsilon$  then ▷ Step is practically zero (stationary point for current  $\mathcal{W}$ )
9:     if all  $\lambda_i \geq 0$  for  $i \in \mathcal{W}$  then ▷ Optimality test
10:        $\text{OptFound} \leftarrow \text{True}$ 
11:        $x_{opt} \leftarrow x$ 
12:       return  $x_{opt}$ 
13:     else
14:        $j_{\text{leave}} \leftarrow \arg \min_{i \in \mathcal{W}} (\lambda_i)$  ▷ Find index of the most negative multiplier
15:        $\mathcal{W} \leftarrow \mathcal{W} \setminus \{j_{\text{leave}}\}$  ▷ Drop constraint from the working set
16:     end if
17:   else ▷ A non-zero search direction is found
18:      $\text{Min}\theta \leftarrow \infty$ 
19:      $j_{\text{block}} \leftarrow -1$ 
20:     for  $i = 1 \rightarrow m$  do ▷ Calculate maximum feasible step length  $\theta$ 
21:       if  $i \notin \mathcal{W}$  and  $a_i^T p > 0$  then
22:          $\theta \leftarrow (b_i - a_i^T x) / (a_i^T p)$ 
23:         if  $\theta < \text{Min}\theta$  then
24:            $\text{Min}\theta \leftarrow \theta$ 
25:            $j_{\text{block}} \leftarrow i$  ▷ Keep track of the blocking constraint
26:         end if
27:       end if
28:     end for
29:     if  $\text{Min}\theta \geq 1$  then
30:        $x \leftarrow x + p$  ▷ Full Newton step
31:     else
32:        $x \leftarrow x + \text{Min}\theta \cdot p$  ▷ Partial step to the boundary
33:        $\mathcal{W} \leftarrow \mathcal{W} \cup \{j_{\text{block}}\}$  ▷ Add blocking constraint to working set
34:     end if
35:   end if
36: end while
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